

Resolution: simplifying a conclusion

Suppose $p \vee q \vee r$ is true and each of p , q , and r implies s . Rewrite the implications as disjunctions, then resolve:

1. $p \vee q \vee r$ premise
2. $\neg p \vee s$ from $p \rightarrow s$
3. $\neg q \vee s$ from $q \rightarrow s$
4. $\neg r \vee s$ from $r \rightarrow s$
5. $q \vee r \vee s$ resolution of 1 and 2
6. $r \vee s$ resolution of 5 and 3
7. $s \vee s$ resolution of 6 and 4
8. s idempotent law.

Another resolution example

Simplify $(p \vee q) \wedge (\neg p \vee q)$:

1. $p \vee q$ from the conjunction
2. $\neg p \vee q$ from the conjunction
3. $q \vee q$ resolution of 1 and 2
4. q idempotent law.

Exhaustive proof

An exhaustive proof checks every possible case of a proposition. If the proposition holds in every case, it is proved.

Proof by cases

Use proof by cases when one argument does not cover every possibility. Divide the possibilities into cases and prove the claim in each case.

Without loss of generality

When cases are symmetric, it can be enough to prove one representative case and state that the others follow by the same argument. This is called proving the claim **without loss of generality** (WLOG).